

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1563

Springfield Power Llc 138 kV
April 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1563. The Interconnection Customer proposes a 231.7 MW solar photovoltaic facility interconnecting at the Springfield Power Llc 138 kV substation in the Virginia area. The request was received on September 01, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 459 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,256,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
INTERNATIONAL PAPER VALLIANT OK - WEST WINCHESTER 138.0 kV L	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV L	129.5	427	7.3
SANTA ROSA ENERGY CENTER - WEST WINCHESTER 138.0 kV Circuit 1	W WINCHESTER - SANTA ROSA ENERGY CENTER 138.0 kV Line	127.7	303	5.1
WEST WINCHESTER - SANTA ROSA ENERGY CENTER 138.0 kV	W WINCHESTER - SANTA ROSA ENERGY CENTER 138.0 kV	127.7	303	5.1
MIDSET - BRIDGE RIVER 138.0 kV	ABBOTSFORD - BRIDGE RIVER 138.0 kV	125.3	210	36.1
PLANT CITY - CEDARBURG S 138.0 kV L	EDINBURG - SPRINGFIELD POWER LLC SUBSTATION 138.0 kV Line	122.4	204	31.3
BRIDGE RIV - TES FILER CITY STATION 138.0 kV	URBAN - AMERICAN 345.0 kV	120.7	371	42.3
CLARK COUNTY 345/138kV Transformer Bank	ARTESIA COUNTRY CLUB - CLARK COUNTY 345.0 kV Line	119.6	512	6.8
PAGE - PURDY 138.0 kV L	MITCHELL GARDENS - KAPSTONE 138.0 kV Line	114.6	586	8.5
DUNN - CEDARBURG SOUTH 138.0 kV	TES FILER CITY STA - BRIDGE RIV 138.0 kV	113.0	243	46.8
FORT BENTON - DETOUR 138.0 kV	DETOUR - SILOAM SPRINGS 138.0 kV	108.3	424	6.2
GLEN HEAD - DUNN 138.0 kV Line	TES FILER CITYSTATION - BRIDGE RIVER 138.0 kV	107.8	252	46.8
CELERON - YELM PP 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	104.9	376	5.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
KILLINGSWORTH - YELM PP 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV	104.8	365	5.1
TES FILER CITY STATION - EDINBURG 138.0 kV	PAGE - PURDY 138.0 kV L	104.5	399	57.2
CELERON - SANDALFOOT 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV Line	104.2	369	6.0
SPRINGFIELD POWER LLC - EDINBURG SUBSTATION 138.0 kV L	PAGE - PURDY 138.0 kV	104.1	412	57.2
NORTH MESA - SANDALFOOT 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	104.0	355	6.0
COLUMBUS CITY 138-1 kV Autotransformer	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV L	102.7	495	5.4
LOUGHRIDGE - GLEN HEAD 138.0 kV	BRIDGERIV - TES FILER CITY STATION SUBSTATION 138.0 kV	102.5	231	15.5
LURAY HYDRO STA - PUDDLEDOK RD 138.0 kV Line	PURDY SUBSTATION - CHATOM 138.0 kV	100.8	327	9.1
PUDDLEDOK RD - MOUNT JACKSON SUBSTATION 138.0 kV	PURDY - CHATOM SUBSTATION 138.0 kV	100.8	334	9.1
PURDY - LURAY HYDRO STATION 138.0 kV Line	PURDY - CHATOM 138.0 kV	100.7	328	9.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
INTERNATIONAL PAPER VALLIANT OK - WEST WINCHESTER 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,306,000	24% (\$562,000)
SANTA ROSA ENERGY CENTER - WEST WINCHESTER 138.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$5,550,000	30% (\$1,673,000)
WEST WINCHESTER - SANTA ROSA ENERGY CTR 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,737,000	30% (\$1,730,000)
MIDSET - BRIDGE RIV 138.0 kV	Rebuild line with higher-capacity conductor	\$6,372,000	45% (\$2,886,000)
PLANT CITY - CEDARBURG S 138.0 kV L	Rebuild line with higher-capacity conductor	\$14,255,000	35% (\$5,000,000)
TES FILER CITY STA - BRIDGE RIVER 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,646,000	37% (\$1,734,000)
CLARK COUNTY 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$15,329,000	14% (\$2,138,000)
PAGE - PURDY 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,557,000	7% (\$237,000)
DUNN - CEDARBURG SOUTH 138.0 kV	Rebuild line with higher-capacity conductor	\$9,241,000	40% (\$3,666,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FORT BENTON - DETOUR 138.0 kV	Reconductor line	\$3,125,000	7% (\$221,000)
GLEN HEAD - DUNN 138.0 kV Line	Reconductor line	\$2,111,000	40% (\$837,000)
CELERON - YELM PP 138.0 kV Line	Reconductor line	\$1,831,000	71% (\$1,293,000)
YELM PP - KILLINGSWORTH 138.0 kV	Reconductor line	\$639,000	71% (\$451,000)
TES FILER CITY STATION - EDINBURG 138.0 kV line	Reconductor line	\$2,285,000	44% (\$1,010,000)
CELERON - SANDALFOOT 138.0 kV	Reconductor line	\$814,000	32% (\$264,000)
SPRINGFIELD POWER LLC - EDINBURG SUBSTATION 138.0 kV L	Reconductor line	\$2,345,000	44% (\$1,037,000)
NORTH MESA - SANDALFOOT 138.0 kV	Reconductor line	\$663,000	32% (\$215,000)
COLUMBUS CITY 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,709,000	25% (\$1,172,000)
LOUGHRIDGE - GLEN HEAD 138.0 kV	Reconductor line	\$1,643,000	64% (\$1,051,000)
PUDDLEDOCK RD - LURAY HYDRO STATION 138.0 kV	Reconductor line	\$3,806,000	14% (\$517,000)
PUDDLEDOCK RD - MOUNT JACKSON SUBSTATION 138.0 kV	Reconductor line	\$2,668,000	14% (\$362,000)
PURDY - LURAY HYDRO STATION 138.0 kV Line	Reconductor line	\$1,475,000	14% (\$200,000)
	Total allocated to project		\$28,256,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.